

AWS Storage

Module 4 — AWS Technical Essentials

Three Storage Types

Block

- Like a hard drive
- Low latency, high IOPS
- Databases, boot volumes

File

- Shared filesystem (NFS/SMB)
- Multiple instances access
- EFS, FSx

Object

- Flat namespace + metadata
- Massive scalability
- Backups, media, data lakes

⚡ Instance Store vs EBS

Instance Store

- Physically attached to host
- Ephemeral (data lost on stop)
- Highest possible IOPS
- Use for caches, temp data

Amazon EBS

- Network-attached volumes
- Persistent across stop/start
- Snapshots for backup
- Use for databases, OS volumes

💾 EBS Volume Types

SSD-Backed

- **gp3/gp2:** general purpose (boot, apps)
- **io2/io1:** high-performance databases

HDD-Backed

- **st1:** throughput-optimized (big data)
- **sc1:** cold storage (infrequent access)

Default choice: gp3 for most workloads (balanced cost and performance).

📦 What Is Amazon S3?

Simple Storage Service — store and retrieve any amount of data, anytime, from anywhere.

- Objects stored in buckets (globally unique name)
- 11 nines (99.999999999%) durability
- Access via HTTPS / REST API
- Up to 5 TB per object

💰 S3 Storage Classes

- **S3 Standard:** frequent access, lowest latency
- **S3 Intelligent-Tiering:** auto-moves between tiers
- **S3 Standard-IA:** infrequent access, lower cost
- **S3 Glacier / Deep Archive:** archival, minutes to hours retrieval

Tip: Use lifecycle policies to automatically transition objects between classes.

🔒 S3 Features

- **Versioning:** keep multiple versions of objects
- **Encryption:** SSE-S3, SSE-KMS, or client-side
- **Access control:** IAM policies, bucket policies, ACLs
- **Block Public Access:** account-level safety net

🎯 Choosing the Right Storage

Use EBS when...

- Databases need low latency
- Boot volumes for EC2
- Single-instance block access

Use S3 when...

- Storing files, images, backups
- Data lakes and analytics
- Static website hosting

🛠️ Lab Time

Create an S3 Bucket for the App

⏱️ 20 minutes

- Create an S3 bucket for employee photos
- Configure bucket policy and permissions
- Upload objects and verify access